

- RAG Learning

Core Practicals (8 mandatory)

Edexcel GCSE Biology (1BIO) — Core Practicals for Papers 1 & 2

Core Practicals (CPs) are the eight mandatory experiments specified by Edexcel, numbered by their specification point (1.6, 1.10, 1.13B, 1.16, 5.18B, 6.5, 8.11, 9.5). Questions about these appear on both Paper 1 and Paper 2. You must be able to describe the *method*, identify *variables*, interpret *results*, evaluate *errors*, and suggest *improvements*. Roughly 15–20% of the written papers test practical skills.

For every practical, remember the scientific-method spine: **Aim → Hypothesis → Variables (IV, DV, CVs) → Method → Results → Analysis → Evaluation**. Markschemes reward precision (units on axes, controlling a named variable, stating *how* you kept something constant — not just "kept it the same").

CP 1.6 — Investigating Specimens with a Light Microscope (Magnification & Drawings)

Aim

To use a light microscope with an eyepiece graticule (and stage micrometer for calibration) to observe and measure the size of plant cells (onion epidermis, stained with iodine) and animal cells (human cheek cells, stained with methylene blue), and to calculate magnification using the formula $\text{magnification} = \text{image size} \div \text{actual size}$.

Equipment / apparatus

- Light microscope with ×4, ×10, ×40 objective lenses and ×10 eyepiece
- Glass microscope slides and cover slips

- Mounted needle, forceps, scalpel, white tile
- Onion (for onion cells) and sterile cotton buds (for cheek cells)
- Iodine solution (for onion) and methylene blue (for cheek cells)
- Paper towel / filter paper for removing excess stain
- Eyepiece graticule and stage micrometer (for calibration)
- Lamp or built-in illuminator

Method (onion cells)

1. Peel a thin layer of epidermis from the inner surface of an onion scale using forceps.
2. Place the tissue flat on a slide — do **not** let it fold over.
3. Add one drop of iodine solution to stain the nuclei and cell walls.
4. Lower a cover slip at an angle using a mounted needle to avoid trapping air bubbles.
5. Blot away excess stain with filter paper.
6. Place the slide on the stage, clip it in, and start on the lowest objective ($\times 4$).
7. Use the coarse focus knob to bring cells roughly into view, then the fine focus knob.
8. Switch to $\times 10$ and then $\times 40$, refocusing with the fine knob only.
9. Draw a labelled scientific drawing: sharp lines, no shading, title, magnification stated.

Method (cheek cells)

1. Rub the inside of the cheek with a sterile cotton bud.
2. Smear the bud onto a slide and add one drop of methylene blue.
3. Add a cover slip as above and observe on $\times 10$ then $\times 40$.
4. Dispose of the cotton bud in disinfectant immediately.

Variables

- **Independent variable:** cell type (plant vs animal) or objective lens magnification.
- **Dependent variable:** observed cell size / number of graticule divisions spanned.
- **Control variables:** stain volume, illumination, sample thickness, same microscope / graticule.

Results / expected observations

- **Onion cells:** brick-shaped, closely packed, clear cell wall, nucleus visible as a darker dot, large central vacuole. No chloroplasts in the epidermis.
- **Cheek cells:** round / irregular, no cell wall, central dark-blue nucleus, thin membrane, scattered — not tessellated.

Calculate magnification: for a cell drawn 40 mm long on paper that measures 0.1 mm (100 μm)
actual $\rightarrow$ magnification = $40 \div 0.1 = \times 400$.

Calibrating the graticule: align the eyepiece graticule with the stage micrometer. If 100 graticule divisions = 1 mm at $\times 10$, each division = 10 μm . At $\times 40$ each division will be 2.5 μm .

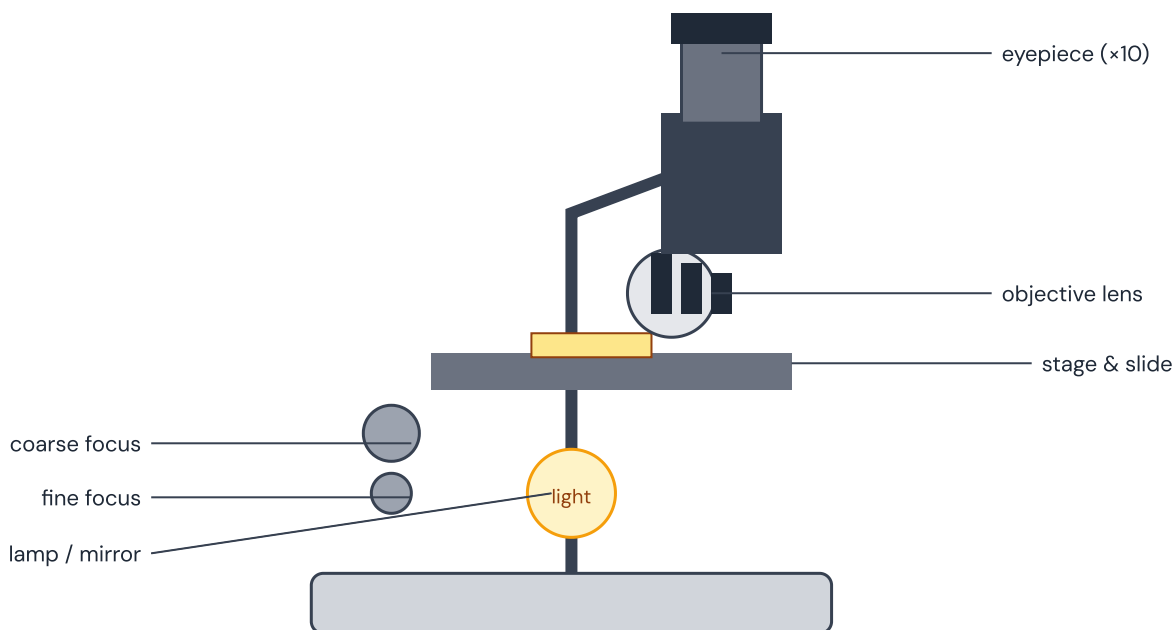
Safety / risk assessment

- Iodine stains skin and clothes — wear a lab coat and wash hands.
- Methylene blue is an irritant — avoid skin contact.
- Glass slides and scalpels are sharp — handle carefully, dispose in sharps bin.
- Cheek cells are biohazardous — *only use your own* cotton bud, and dispose in disinfectant.

Improvements / sources of error

- Air bubbles trapped under cover slip $\rightarrow$ lower at an angle.
- Sample too thick $\rightarrow$ use a thinner layer (single cell thickness).
- Uneven staining $\rightarrow$ leave stain to act for 30 s before blotting.
- Parallax error when counting graticule divisions $\rightarrow$ view from directly above.

CP 1.6 — Light Microscope



CP 1.10 — Effect of pH on Enzyme (Amylase) Activity

Aim

To investigate how **pH** affects the rate at which amylase breaks down starch, using iodine solution as an indicator. The **rate** of reaction is calculated as $1 \div \text{time taken (s}^{-1}\text{)}$ for the iodine colour to stop changing from blue-black to orange/yellow.

Equipment / apparatus

- 1% amylase solution; 1% starch solution
- pH buffers: pH 3, 5, 7, 9, 11
- Iodine solution in a dropping pipette
- White spotting tile (for iodine tests)
- Test tubes, stopwatch, pipettes, thermometer
- Water bath at 35 °C — Edexcel specifies a beaker of water heated with a **Bunsen burner** (on a tripod and gauze) to hold the temperature steady

Method

1. Place one drop of iodine solution into each well of a spotting tile.
2. Add 2 cm³ of starch solution and 1 cm³ of pH 3 buffer to a test tube.
3. Place the tube in a 35 °C water bath for 5 minutes to equilibrate.
4. Add 2 cm³ of amylase solution and **start the stopwatch**. Mix.
5. Every 30 s (continuous sampling), remove a drop of the mixture with a clean pipette and add it to a fresh well of iodine on the spotting tile.
6. Record the time at which the iodine **no longer turns blue-black** (stays orange) — starch has been fully broken down.
7. Repeat for pH 5, 7, 9, and 11. Do three trials at each pH and take a mean.
8. Calculate rate = $1 \div \text{time (s}^{-1}\text{)}$ and plot against pH.

Variables

- **Independent variable:** pH of buffer.
- **Dependent variable:** time for starch to be fully digested (rate = 1/time).
- **Control variables:** temperature (35 °C), amylase concentration, amylase volume, starch concentration, starch volume.

Results / expected observations

- Fastest reaction (shortest time) at **pH 7** — the optimum for salivary amylase.
- Rate falls either side of pH 7; at pH 3 and pH 11 the enzyme is **denatured** — the active site changes shape and can no longer bind starch.
- Graph of rate vs pH is a classic bell curve peaking at pH 7.

Safety / risk assessment

- Iodine stains and is an irritant — wear goggles.
- Acid/alkali buffers — low hazard at these dilutions but wash spills off skin.
- Hot water bath — handle tubes with tongs.

Improvements / sources of error

- 30 s intervals are imprecise → sample every 10 s near the end point.
- Human judgement of colour change is subjective → use a **colorimeter** for objective data.
- Reaction starts before water bath equilibration → warm reagents separately first.
- Cross-contamination of iodine drops → use a fresh pipette for each sample.

CP 1.13B — Food Tests: Starch, Reducing Sugars, Proteins and Fats

Aim

To use chemical reagents to identify which biological molecules are present in a food sample: **starch** (iodine), **reducing sugars** (Benedict's solution, heated), **protein** (Biuret test) and **fats/oils** (emulsion test). These are **qualitative** tests — they show whether a molecule is present, not exactly how much. (1.13B is a separate-science point, so it is in Biology 1BIO but not Combined Science.)

Equipment / apparatus

- Food samples (e.g. bread, milk, egg white, cooking oil, glucose solution), pestle and mortar, distilled water
- Filter funnel and filter paper, test tubes and rack, spatula, pipettes, measuring cylinder
- Iodine solution; Benedict's solution; Biuret reagents (potassium hydroxide solution and dilute copper sulfate solution); ethanol
- Water bath (beaker of water, ~80 °C or boiling, heated by a Bunsen or kettle), test-tube holder
- Spotting tile (for the starch test on solids)

Method

1. **Prepare the sample:** grind the food with a little distilled water using a pestle and mortar, then filter to get a clear solution. Divide it between labelled test tubes (keep a small solid portion for the fat test).
2. **Starch:** add a few drops of iodine solution. Orange-brown → **blue-black** = starch present.

3. **Reducing sugars:** add an equal volume (about 2 cm³) of Benedict's solution and stand the tube in a hot water bath for about 5 minutes. Blue → green → yellow → orange → **brick-red** precipitate = reducing sugar present; the further along this sequence, the more sugar.
4. **Protein (Biuret test):** add about 2 cm³ of potassium hydroxide solution, then a few drops of copper sulfate solution and shake gently. Blue → **purple/lilac** = protein present.
5. **Fats (emulsion test):** shake a small amount of the food with about 2 cm³ of ethanol so any fat dissolves, then pour the ethanol into a tube of water. A **cloudy white emulsion** = fat present.
6. Run each test on a tube of distilled water as a **negative control** so you can compare colours.

Variables

- **Independent variable:** type of food tested.
- **Dependent variable:** colour change (presence or absence of each molecule).
- **Control variables:** volume of sample and reagent, heating time and water-bath temperature (Benedict's), same concentration of reagents, clean apparatus for each test.

Results / expected observations

Test	Reagent	Negative result	Positive result
Starch	Iodine solution	Orange-brown	Blue-black
Reducing sugar (e.g. glucose)	Benedict's + heat	Stays blue	Green / yellow / orange / brick-red
Protein	Biuret (KOH + CuSO ₄)	Blue	Purple / lilac
Fat / oil	Ethanol, then water	Clear	Cloudy white emulsion

Sucrose (table sugar) is **not** a reducing sugar, so it gives a negative Benedict's result.

Safety / risk assessment

- Ethanol is highly **flammable** — keep it away from any Bunsen flame; heat Benedict's tubes in a water bath, never directly.
- Potassium hydroxide is corrosive/irritant and copper sulfate is harmful — wear goggles and wash off any skin contact.
- Hot water bath — use a test-tube holder; point tubes away from people.
- Do not eat any food used in the lab; check for allergies (e.g. nuts, milk).

Improvements / sources of error

- Colour judgements are subjective → compare against a control and a known positive, or use a colorimeter for a semi-quantitative Benedict's result.
- Coloured food solutions can mask the result → filter well and dilute if needed.
- Heating time/temperature varies between tubes → heat all Benedict's tubes together in the same bath for the same time.
- Contaminated pipettes give false positives → use a clean pipette for each reagent and sample.

CP 1.16 — Investigating Osmosis in Potatoes

Aim

To investigate how the mass of potato cylinders changes when they are placed in sucrose solutions of different concentrations, and to determine the concentration at which there is **no mass change** — which equals the concentration of the cell sap (isotonic point).

Equipment / apparatus

- Potato, cork borer, tile, scalpel, ruler
- Distilled water and sucrose solutions: 0.0, 0.2, 0.4, 0.6, 0.8, 1.0 mol/dm³
- Six boiling tubes in a rack, labels
- Electronic balance (to 2 d.p.), paper towels

- Stopwatch, measuring cylinder

Method

1. Use a cork borer to cut several cylinders of potato, then trim them to the same length (e.g. 3 cm) with a scalpel.
2. Blot each cylinder gently with paper towel to remove surface water.
3. Record the **initial mass** of each cylinder on a balance.
4. Place one cylinder into each boiling tube containing 20 cm³ of sucrose solution of a different concentration.
5. Leave for at least 30 minutes (ideally an hour) at room temperature.
6. Remove, blot, and record the **final mass**.
7. Calculate **percentage change in mass**: $((\text{final} - \text{initial}) \div \text{initial}) \times 100$.
8. Plot % change in mass (y-axis) against sucrose concentration (x-axis). Draw a smooth best-fit line; where it crosses the x-axis = isotonic concentration.

Variables

- **Independent variable:** sucrose concentration (mol/dm³).
- **Dependent variable:** percentage change in mass of potato cylinder.
- **Control variables:** volume of solution, length / surface area of potato, time in solution, temperature, blotting procedure, same potato (same variety).

Results / expected observations

Sucrose conc. (mol/dm ³)	% change in mass	Interpretation
0.0 (water)	+15 to +20%	Water enters by osmosis; cells turgid
0.2	+5%	Still hypotonic to cell sap
0.3–0.4	~0%	Isotonic — equals cell sap concentration
0.6	-8%	Water leaves cells; mass falls

1.0	-20%	Plasmolysis — cells flaccid
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Expected graph shape: a straight line with negative gradient starting in the positive region, crossing the x-axis between 0.3 and 0.4 mol/dm³, continuing into the negative region.

Safety / risk assessment

- Scalpels and cork borers are sharp — always cut onto a tile, cutting away from fingers.
- Sucrose is low hazard but may spill — wipe up and wash hands.
- Wear safety goggles.

Improvements / sources of error

- Inconsistent blotting → standardise (same person, same number of dabs).
- Potato cylinders of different sizes → cut all to the same length *and* use the same borer.
- Evaporation of solution → cover tubes with parafilm.
- Short time → equilibrium not reached; run for at least an hour.
- Repeat and take a mean to spot anomalies.

CP 5.18B — Effect of Antiseptics, Antibiotics or Plant Extracts on Microbial Cultures

Aim

To use **aseptic technique** to grow bacteria on an agar plate and investigate the antibacterial effect of different antiseptics, antibiotics, or plant extracts (e.g. garlic, mint) by measuring the **zone of inhibition** (clear area) around paper discs soaked in each substance. Larger zone = more effective inhibitor.

Equipment / apparatus

- Sterile nutrient agar plates (Petri dishes)
- Broth culture of a safe bacterium (e.g. *Bacillus subtilis*, *Micrococcus luteus*)

- Sterile pipette, sterile spreader (or cotton swab)
- Filter-paper discs (sterile), forceps flamed in ethanol
- Antiseptics / plant extracts / antibiotic discs
- Bunsen burner, autoclave tape, incubator (25 °C for school safety)
- Ruler, marker pen

Method

1. Sterilise the bench with disinfectant. Work near a lit Bunsen burner — its updraft carries airborne microbes up and away from the open dish.
2. Pipette 0.1 cm³ of bacterial broth onto the agar and spread it evenly with a sterile spreader — work close to the flame; lift the lid only a little.
3. Using flamed forceps (dipped in ethanol, passed through flame, cooled), place paper discs soaked in each test substance onto the agar, evenly spaced. Include a disc soaked in sterile water as a **negative control**.
4. Label the **base** of the dish (not the lid, which could be swapped), writing around the edge, with your name, date and disc identity.
5. Secure the lid with two pieces of tape (do NOT seal all the way round — anaerobic conditions could let harmful bacteria grow).
6. Incubate upside down at 25 °C for 24–48 hours.
7. After incubation, measure the **diameter** of each clear zone (zone of inhibition) with a ruler, through the lid.
8. Calculate the **area** of each zone: $A = \pi r^2$. Compare substances.

Variables

- **Independent variable:** type of antiseptic / plant extract / antibiotic.
- **Dependent variable:** diameter (and/or area) of the zone of inhibition.
- **Control variables:** bacterial species, volume / concentration of bacteria, disc size, volume of substance on each disc, agar type, incubation time and temperature.

Results / expected observations

- Strong antibiotic (e.g. penicillin on susceptible strain): large clear zone (> 20 mm).
- Antiseptics (ethanol, bleach): moderate zones.
- Plant extracts (garlic): small but measurable zone.
- Water control: no zone — bacterial lawn grows up to the disc edge.

Safety / risk assessment

- Use **aseptic technique** throughout to prevent contamination.
- Incubate at $\leq 25\text{ }^{\circ}\text{C}$ in schools — prevents growth of human pathogens that prefer $37\text{ }^{\circ}\text{C}$.
- Do **not** open plates after incubation — dispose sealed.
- Autoclave (pressure steriliser) all plates after use.
- Wash hands; disinfect bench before and after.
- Bunsen flame — tie hair back, clear flammable materials.

Improvements / sources of error

- Uneven bacterial spread → plate appears patchy; spread more thoroughly.
- Different volumes of substance on discs → use a fixed micropipette (e.g. 10 μl each).
- Zones overlap → use larger plates or fewer discs.
- Measuring diameter only → area (πr^2) is more representative because a zone that is twice the diameter has *four* times the area.
- Repeat plates and take a mean — discard contaminated plates.

CP 6.5 — Effect of Light Intensity on the Rate of Photosynthesis

Aim

To investigate how **light intensity** (varied by changing the distance between a lamp and pondweed) affects the rate of photosynthesis, measured as the number of oxygen bubbles released per minute by *Cabomba* or *Elodea*. Edexcel's own guidance uses **algal balls** (algae immobilised in alginate) in hydrogencarbonate indicator placed at different distances from the lamp, timing the colour change — the spec requires algal balls "or similar", with rate measured

against distance from the light, so pondweed is an accepted alternative. Alternative: the leaf-disc buoyancy method (discs rise as O_2 is produced).

Equipment / apparatus

- Fresh sprig of *Elodea* / *Cabomba* pondweed
- Boiling tube / beaker of sodium hydrogen carbonate solution ($NaHCO_3$, ~1%) as a CO_2 source
- Bench lamp, ruler or metre rule
- Stopwatch, thermometer
- Heat-shield (beaker of water) between lamp and plant to prevent warming

Method

1. Cut ~8 cm of pondweed, recut the stem underwater at an angle to aid bubble release.
2. Place the sprig in a boiling tube of $NaHCO_3$ solution, cut end upwards.
3. Put the tube in a water-filled beaker (heat shield). Record the water temperature.
4. Position a lamp 10 cm from the tube. Let the plant acclimatise for 5 minutes.
5. Count the number of bubbles released per minute. Repeat 3 times; take the mean.
6. Move the lamp to 20, 30, 40, 50 cm. Repeat steps 4–5 at each distance.
7. Calculate light intensity using the **inverse-square law**: $I \propto 1 \div d^2$ (where d is distance in metres).
8. Plot rate (bubbles/min) against $1/d^2$.

Variables

- **Independent variable:** distance from lamp (converted to light intensity $1/d^2$).
- **Dependent variable:** rate of photosynthesis (bubbles per minute).
- **Control variables:** temperature (using heat shield), CO_2 concentration ($NaHCO_3$ solution), same pondweed sprig, same wavelength / lamp, same acclimatisation time.

Results / expected observations

- As distance decreases (light intensity increases), bubble rate **increases**.

- At high intensities the rate levels off — another factor (CO_2 , temperature) becomes **limiting**.
- Graph vs $1/d^2$ is approximately linear then plateaus.

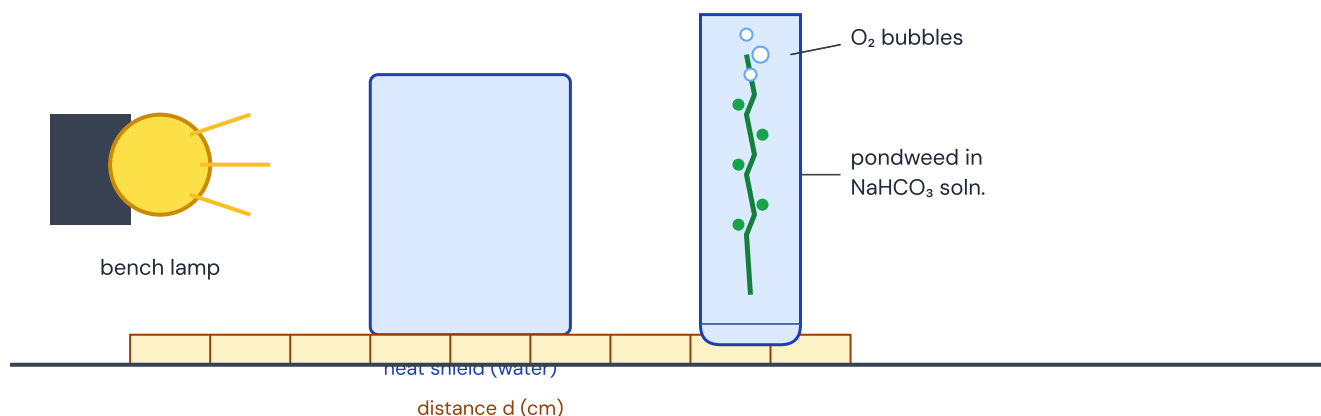
Safety / risk assessment

- Lamp gets hot — do not touch; use a heat shield.
- Electrical apparatus + water — keep cables away from spills.
- Glass tubes — clamp securely.

Improvements / sources of error

- Bubbles vary in size → collect gas in an inverted measuring cylinder / gas syringe for **volume per minute**.
- Background room light → work in a darkened room.
- Plant produces O_2 more slowly when stressed → give 5 min acclimatisation at each new intensity.
- With algal balls, compare the indicator colour against a set of standard colours (or use a colorimeter) at fixed times rather than judging by eye.
- Use the **leaf-disc buoyancy method**: punch 10 discs from a leaf, remove air by vacuum, drop into NaHCO_3 solution, count how many rise in each 30 s interval.

CP 6.5 — Pondweed & Lamp Setup



CP 8.11 — Rate of Respiration in Living Organisms

Aim

To measure the **rate of respiration** of small living organisms (e.g. woodlice, maggots or germinating peas) using a simple respirometer, and to investigate how **temperature** (set with a water bath) affects it. Soda lime absorbs CO_2 produced, so the volume of gas change equals the volume of O_2 consumed. Movement of a coloured manometer fluid measures O_2 uptake. A boiled-seed control shows that the effect requires living organisms.

Equipment / apparatus

- Respirometer tube with capillary side-arm + manometer fluid
- Germinating peas (live) and boiled peas (dead control)
- Soda lime (to absorb CO_2)
- Gauze platform / cotton wool (to keep peas above soda lime)
- Water bath at fixed temperature (e.g. $20\text{ }^{\circ}\text{C}$)
- Stopwatch, ruler, syringe for resetting bubble

Method

1. Place soda lime in the bottom of the respirometer tube. Cover with gauze/cotton wool.
2. Add germinating peas on top of the gauze.
3. Seal the tube with a bung connected to a capillary tube containing a coloured liquid droplet (manometer).
4. Submerge the apparatus in a water bath to keep temperature constant. Leave 5 min to equilibrate.
5. Open the tap briefly to equalise pressure, then close.
6. Record the position of the coloured droplet every minute for 10 minutes.
7. Calculate the distance moved per minute; using the capillary bore, convert to **volume of O_2 consumed per minute per gram of peas** ($\text{mm}^3\text{ min}^{-1}\text{ g}^{-1}$).

8. Repeat with the water bath at several different temperatures (e.g. 15, 20, 25, 30 °C), equilibrating each time.
9. Repeat with **boiled (dead) peas** as a control — droplet should not move. Repeat with an empty tube (no peas).

Variables

- **Independent variable:** temperature of the water bath (the Edexcel version); live vs boiled peas / empty tube act as controls.
- **Dependent variable:** rate of O₂ uptake (distance moved by droplet ÷ time).
- **Control variables:** mass and type of organisms, volume of soda lime, apparatus geometry, equilibration time (temperature is the independent variable, so each run is held at one set water-bath temperature).

Results / expected observations

- Live germinating peas: droplet moves steadily **towards the peas** as O₂ is used up.
- Boiled peas: no movement — confirms respiration requires living cells.
- Empty tube: no movement — confirms the apparatus is sealed.
- Rate typically doubles for every 10 °C rise ($Q_{10} \approx 2$), up to ~40 °C, then falls as enzymes denature.

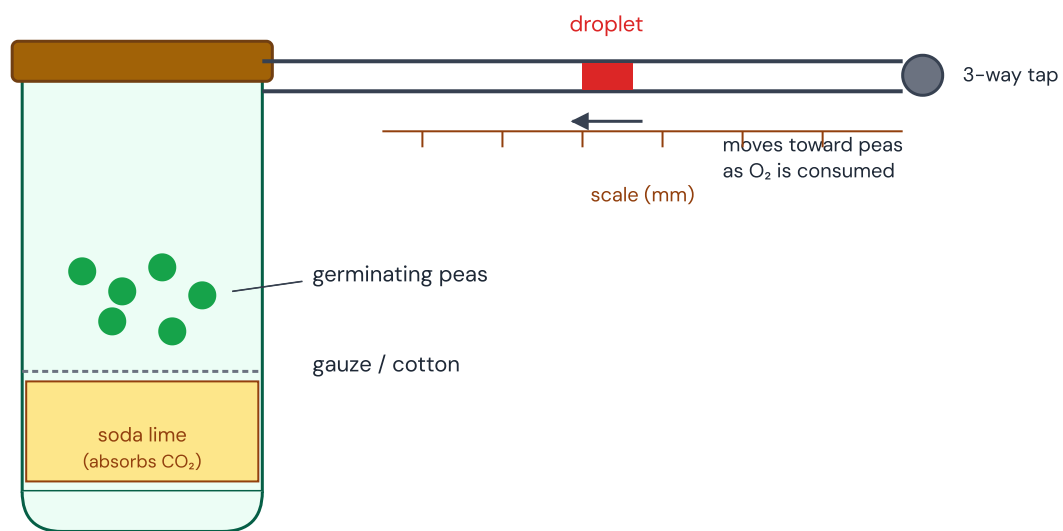
Safety / risk assessment

- Soda lime is **corrosive** — wear goggles and gloves; avoid contact with skin.
- Keep soda lime away from peas (use gauze); direct contact would damage the seeds.
- Glass capillary tubing is fragile — handle carefully.
- Wash hands after handling peas or invertebrates.
- **Ethics:** if using animals (woodlice, maggots), handle gently, keep them out of contact with the soda lime, limit the time in the tube and the temperature range, and return them to their habitat afterwards.

Improvements / sources of error

- Pressure / temperature changes in the room move the droplet → use a water bath and (beyond GCSE) a thermobarometer (second tube with glass beads instead of peas, connected in parallel) to correct.
- Leaks → test by pressing bung; droplet should return.
- Different masses of peas → divide rate by mass to get rate per gram.
- Repeat and mean.

CP 8.11 — Simple Respirometer



CP 9.5 — Organisms and Their Environment: Quadrats and Belt Transects

Aim

To estimate the population size of a plant species in a habitat using random quadrats, **and** (the focus of the Edexcel practical) to investigate how the distribution of organisms changes along an environmental gradient (e.g. light intensity under a tree) using a belt transect.

Equipment / apparatus

- 0.5 m × 0.5 m (0.25 m²) wire quadrat frame, or 1 m × 1 m
- Two 20 m measuring tapes (to set up grid coordinates)

- Random number generator (calculator / phone)
- Identification keys for plants
- Light meter (lux), soil thermometer, moisture meter (for environmental factors)

Method (random sampling — estimating population)

1. Lay two tapes at right angles to form x- and y-axes of a grid on the field (e.g. 20 m × 20 m).
2. Use random numbers to generate coordinate pairs (e.g. 7, 14).
3. Place a 0.25 m² quadrat at each coordinate.
4. Count the number of individuals of the target species (e.g. daisies) in each quadrat.
5. Repeat for at least 10 quadrats (more is better).
6. Calculate the mean number per quadrat.
7. Estimate the total population: $\text{mean per quadrat} \times (\text{total area} \div \text{quadrat area})$.
e.g. mean = 5 daisies / 0.25 m²; field = 400 m². Estimated total = 5 × (400 ÷ 0.25) = 8000 daisies.

Method (belt transect — distribution)

1. Run a tape measure in a straight line along the environmental gradient (e.g. from under a tree into open grass).
2. Place a quadrat every 1 m (or at set intervals).
3. At each quadrat, estimate **percentage cover** of each species and measure light intensity (lux).
4. Plot percentage cover and light intensity against distance.

Variables

- **Random sampling IV:** position of quadrat (random). **DV:** number of individuals.
- **Transect IV:** distance along gradient. **DV:** % cover or number of species.
- **Control variables:** same observer, same quadrat size, same time of day, same identification key.

Results / expected observations

- Under a tree: low light, fewer species, shade-tolerant species (moss, ivy) dominate.
- In the open: higher light, more grass and daisies.
- Graph of % cover vs distance shows species replacing each other along the gradient.

Safety / risk assessment

- Work in pairs, stay in sight of the teacher.
- Check ground for glass, hazards, nettles.
- Wash hands after contact with soil.
- Sun safety: hats, sunscreen, water.

Improvements / sources of error

- Too few quadrats → larger sample gives a more reliable mean.
- Identification errors → use a key, or photograph unknown species.
- % cover estimates are subjective → have two observers and take the mean.
- "Random" sampling not truly random → use a random-number generator, not guessing.
- For mobile animals (snails etc.) use **capture-mark-recapture** (not required for 1B10) with the Lincoln index: $N = (n_1 \times n_2) \div n_r$.

Not Core Practicals for 1B10 — Useful Supporting Practicals

The two investigations below are **not** among the eight Edexcel core practicals, but the ideas behind them are in the specification content (calorimetry is spec point 1.14B; factors affecting transpiration are in Topic 6), so exam questions can still ask about them.

Energy content of food by calorimetry (spec 1.14B — not a core practical)

Aim

To determine the **energy content per gram** of different foods (e.g. pasta, crisp, peanut, cracker) by burning a known mass of food under a known volume of water and measuring the temperature rise, using $E = m \times c \times \Delta T$.

Equipment / apparatus

- Boiling tube, clamp stand, mounted needle or bent paper clip
- Measuring cylinder, thermometer (0.1 °C if possible)
- Bunsen burner, heat-proof mat
- Electronic balance (2 d.p.), samples of food

Method

1. Measure 25 cm³ of water into a boiling tube using a measuring cylinder.
2. Clamp the boiling tube at an angle, and record the **initial water temperature**.
3. Mass the food sample on a balance and record.
4. Fix the food onto a mounted needle.
5. Ignite the food in a Bunsen flame and **immediately** hold it under the boiling tube.
6. If the flame goes out, re-light and hold back under until the food will no longer burn.
7. Record the **final water temperature**. Calculate ΔT .
8. Calculate **energy (J)** = $m(\text{water in g}) \times 4.2 \times \Delta T$.
9. Calculate **energy per gram of food** = $\text{energy} \div \text{mass of food burnt (J/g)}$.
10. Repeat with different foods and compare.

Variables

- **Independent variable:** type of food.
- **Dependent variable:** temperature rise of water (ΔT), used to calculate energy per gram.
- **Control variables:** volume of water (25 cm³), starting water temperature, distance of food from tube, same thermometer, same boiling tube.

Results / expected observations

- High-fat foods (peanuts, crisps) release the most energy per gram — typically 20–25 kJ/g (lab values often lower due to heat loss).
- Carbohydrate foods (cracker, pasta) ~15–17 kJ/g.
- Protein foods intermediate.
- Laboratory values are typically **much lower** than textbook values — see "sources of error".

Safety / risk assessment

- Open flame — tie hair back, tuck away ties, wear goggles.
- Food peanut samples — check for nut allergies in the class before starting!
- Hot apparatus — allow to cool before handling.
- Do not lean over the Bunsen burner while lighting food.

Improvements / sources of error

- **Huge heat loss** to the surroundings → lag the boiling tube, or use a copper calorimeter.
- Incomplete combustion → use a pure-oxygen **bomb calorimeter** (industry standard).
- Food burns unevenly / falls off needle → re-ignite promptly.
- Thermometer to only 1 °C → use a digital probe (± 0.1 °C).
- Always repeat and take a mean.

Rate of transpiration using a potometer (topic 6 content — not a core practical)

Aim

To investigate how **environmental conditions** (still air, moving air, high humidity, high temperature, high light) affect the rate of transpiration in a leafy shoot, measured as the rate at which an air bubble moves along the capillary of a potometer.

Equipment / apparatus

- Potometer (apparatus with reservoir, capillary tube with scale, shoot holder)

- Leafy shoot (e.g. geranium, privet)
- Bowl of water, sharp scalpel
- Petroleum jelly (to seal joints)
- Stopwatch, ruler
- Fan (moving air), plastic bag (humidity), lamp (light), heater/warm conditions

Method

1. Cut the shoot under water at an angle to prevent air entering the xylem. Keep it submerged.
2. Set up the potometer fully under water — no bubbles anywhere *except* the one you will introduce.
3. Fit the shoot into the rubber bung and seal all joints with petroleum jelly.
4. Lift the potometer out of the water. Introduce a single air bubble by briefly touching the capillary tip on a paper towel.
5. Allow the shoot to acclimatise for 5 min.
6. Record the distance the bubble travels in a fixed time (e.g. 2 minutes).
7. Reset the bubble using the reservoir tap.
8. Change ONE condition at a time: fan on (moving air), bag over shoot (humid), lamp close (bright light), warm room (high temp). Record each rate.
9. Calculate rate = distance ÷ time (mm/min). If the capillary radius r is known, volume = $\pi r^2 \times$ distance, giving $\text{mm}^3 \text{ min}^{-1}$.

Variables

- **Independent variable:** environmental condition (still/moving air, humidity, temperature, light).
- **Dependent variable:** rate of bubble movement = rate of water uptake $\approx$ transpiration rate.
- **Control variables:** same shoot, same leaf surface area, same time of day, same starting position, sealed joints.

Results / expected observations

Condition	Expected rate	Reason
Still air (control)	Baseline	Reference value
Moving air (fan)	↑ highest	Removes water vapour from leaf surface; steeper concentration gradient
High humidity (bag)	↓ lowest	Shallow water-vapour gradient; less diffusion out of stomata
High temperature	↑	Water molecules move faster; stomata may open
High light	↑	Stomata open wider for photosynthesis; more diffusion

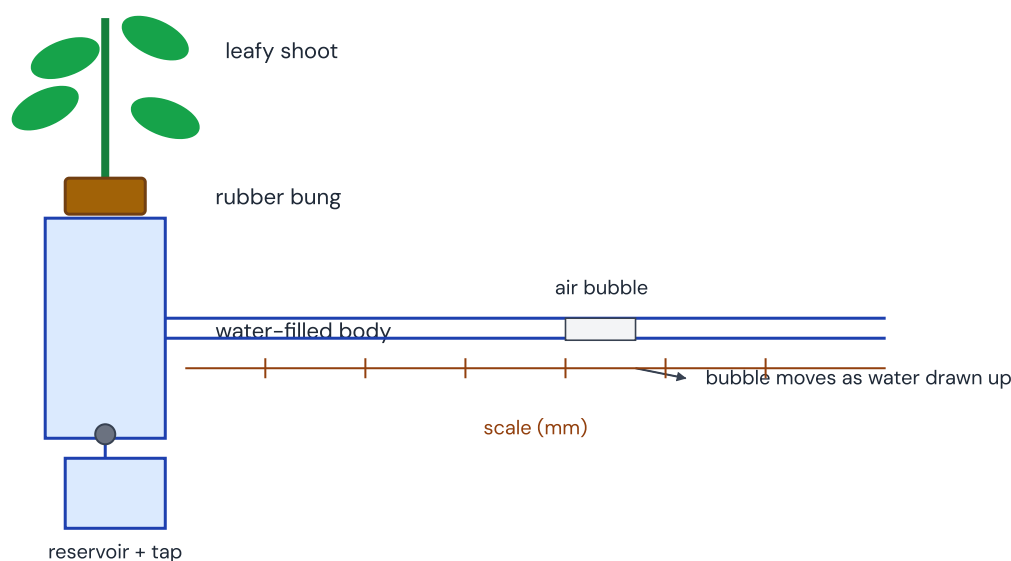
Safety / risk assessment

- Scalpels and glass capillary — sharp; cut away from fingers.
- Electrical fan / lamp near water — keep cables dry.
- Some plants are irritants — wash hands after.

Improvements / sources of error

- Strictly this measures water *uptake*, not transpiration directly; it assumes the two are equal (small error because some water is used in photosynthesis).
- Unsealed joints → air leaks in → reseal with petroleum jelly.
- Leaf area differs between shoots → measure total leaf area and express rate per cm².
- Bubble moves too fast to time → use a shorter interval or a larger-bore capillary.
- Repeat and mean; allow 5 min between conditions.

Potometer (not a core practical)



Practical Skills Summary — Exam Essentials

Exam Tip: Always state IV, DV, and CVs. When an exam question says "plan an investigation", a markscheme almost always awards:

- 1 mark for naming the IV and how it is changed (e.g. "temperature, by using water baths at 10, 20, 30, 40 °C")
- 1 mark for naming the DV and how it is measured (e.g. "rate = $1 \div \text{time}$ for colour change, measured with a stopwatch")
- 1 mark for each named CV and *how* it is controlled (not just "kept the same")
- 1 mark for repeats & mean.

Exam Tip: Accuracy vs Precision. *Accurate* = close to the true value. *Precise* = measurements close to each other (small spread), whether or not they are accurate. An instrument can be precise but inaccurate if it is badly calibrated (e.g. a balance that always reads 0.2 g too heavy).

Exam Tip: Anomalies and repeats. An *anomaly* is a result that doesn't fit the pattern. Identify them by eye or using ± 2 SD. *Exclude* them from the mean, but still record them.

Always do at least **three repeats** to spot anomalies; more for biological variation. Concordant repeats give concordant results — more confidence.

Exam Tip: Units on axes. Every axis must have both a *quantity* and a *unit* in brackets, e.g. "Rate of reaction / s⁻¹". No units = lost marks. Use a sensible scale that uses more than half the paper. Plot points as crosses or dots-in-circles. Draw a best-fit line (or smooth curve), not dot-to-dot.

Exam Tip: Mean vs median vs mode.

- **Mean** — use when data are numerical and roughly symmetrical; sensitive to outliers.
 - **Median** — use when the data are skewed or contain outliers (e.g. length of bean seedlings where one was dead).
 - **Mode** — use for categorical / most-frequent data (e.g. commonest eye colour in class).
- For most core practical calculations (enzyme rates, transpiration, osmosis) you use the **mean**.

Exam Tip: Safety — standard phrases. Wear safety goggles / lab coat / gloves. Tie back hair. Keep flammables away from Bunsen. Dispose of biological material in disinfectant or autoclave. Use a heat shield with lamps. Inform teacher of allergies (e.g. nuts in food tests or calorimetry).

Exam Tip: Evaluation framework. For the 6-mark "evaluate the method" questions, use this structure: (1) *Reliability* — enough repeats, concordant results? (2) *Accuracy* — equipment precision, systematic errors? (3) *Validity* — did you actually measure what you claimed? Were CVs controlled? (4) *Improvements* — propose one specific change per issue.

End of Core Practicals (8 mandatory) — Edexcel GCSE Biology (1BIO).